

Decrypting ASP.NET Identity cookies

Decrypting ASP.NET Identity cookies - @lowleveldesign, Debug notes by Sebastian Solnica.

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Content

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Decrypting ASP.NET Identity cookies

I decided recently I need to learn Python. It's a great scripting language, often used in forensics, diagnostics and debugging tools. There is even a plugin for windbg that allows you to script this debugger in Python language, but it's a subject for another post. Moving back to learning Python – as an exercise I wrote a simple tool to decrypt ASP.NET Identity cookies and ASP.NET Anti-Forgery tokens. You may find it useful in situations when you need to diagnose why one of your users can't sign in into your applications or is not authorize to access one of its parts. It does not perform validation but only decrypts the content using 256-bit AES (let me know in comments if you need some other decryption algorithm to be implemented). Adding validation logic shouldn't be a big deal and the nist library (which I used for cryptographic operations) provides all the necessary functions.

The script goes as follows:

```

# -*- coding: utf-8 -*-
import base64
import logging
import argparse
import struct
import gzip
from StringIO import StringIO
from cryptography.hazmat.primitives.ciphers import Cipher, algorithms, modes
from cryptography.hazmat.primitives import hashes, padding, hmac
from cryptography.hazmat.backends import default_backend

def derivekey(key, label, context, keyLengthInBits):
    lblcnt = 0 if None == label else len(label)
    ctxcnt = 0 if None == context else len(context)
    buffer = ['\x00'] * (4 + lblcnt + 1 + ctxcnt + 4)
    if lblcnt != 0:
        buffer[4:(4 + lblcnt)] = label
    if ctxcnt != 0:
        buffer[(5 + lblcnt):(5 + lblcnt + ctxcnt)] = context
    _writeuint(keyLengthInBits, buffer, 5 + lblcnt + ctxcnt)
    dstoffset = 0
    v = int(keyLengthInBits / 8)
    res = ['\x00'] * v
    num = 1
    while v > 0:
        _writeuint(num, buffer, 0)
        h = hmac.HMAC(key, hashes.SHA512(), backend=default_backend())
        h.update(''.join(buffer))
        hash = h.finalize()
        cnt = min(v, len(hash))
        res[dstoffset:cnt] = hash[0:cnt]
        dstoffset += cnt
        v -= cnt
        num += 1
    return ''.join(res)

def _writeuint(v, buf, offset):
    buf[offset:(offset + 4)] = struct.pack('>I', v)

def _tokendecode(aspnetstr):
    if len(aspnetstr) < 1:
        raise ValueError('Invalid input')

```

```

# add padding if necessary - last character of the string defines the padding length
num = ord(aspnetstr[-1]) - 48
if num < 0 or num > 10:
    return None

return base64.urlsafe_b64decode(aspnetstr[:-1] + num * '=')

def _decode(aspnetstr):
    # add padding if necessary
    pad = 3 - ((len(args.aspnetstr) + 3) % 4)
    if pad != 0:
        aspnetstr += pad * '='
    return base64.urlsafe_b64decode(aspnetstr)

def decrypt(dkey, b):
    # extract initialization vector (256 bit)
    iv = b[0:16]
    decryptor = Cipher(algorithms.AES(dkey), modes.CBC(iv),
        backend=default_backend()).decryptor()
    unpadder = padding.PKCS7(algorithms.AES.block_size).unpadder()

    ciphertext = b[16:-32]
    text_padded = decryptor.update(ciphertext) + decryptor.finalize()
    return unpadder.update(text_padded) + unpadder.finalize()

if __name__ == '__main__':
    # Turn on Logging
    logging.basicConfig(level=logging.DEBUG, format='%(asctime)s %(message)s')

    parser = argparse.ArgumentParser('ASP.NET encryptor/decryptor')
    parser.add_argument('aspnetstr', metavar='aspnet-text', help='ASP.NET encrypted
text')
    parser.add_argument('-skey', required=True, help='Symmetric key for AES
encryption/decryption')
    parser.add_argument('-entype', required=False, help='Type of action that generated
the given encryption text (owinauth or antiforgery)')
    args = parser.parse_args()

    skey = args.skey.decode('hex')
    label = None
    context = None
    compressed = False
    encrypted = None
    if args.entype == 'owinauth':

```

```

        label =
b'>Microsoft.Owin.Security.Cookies.CookieAuthenticationMiddleware\x11ApplicationCookie\x0
2v1'

        context = b'User.MachineKey.Protect'
        compressed = True
        encrypted = _decode(args.aspNetstr)
    elif args.entropy == 'antiforgery':
        label = b'/System.Web.Helpers.AntiXsrf.AntiForgeryToken.v1'
        context = b'User.MachineKey.Protect'
        encrypted = _tokendecode(args.aspNetstr)

    dkey = derivekey(skey, context, label, 256)
    decrypted = decrypt(dkey, encrypted)

    if compressed:
        decrypted = gzip.GzipFile(fileobj=StringIO(decrypted)).read()

    print "%s %s" % (decrypted.encode('hex'), decrypted)

```

It works only with keys explicitly defined in web.config in machineKey section, eg.

```

...
<system.web>
    ...
    <machineKey decryption="AES"
decryptionKey="22d14047d53135334cb08d4b4d7da1dcfccd0eae9e66fea0b8dfdcda085a683"
        validation="HMACSHA256"
validationKey="c2aec26d010bb4224ab2189184cca3c1b43ae9688026ae4a2f851fbf5521c73f" />
    <httpRuntime targetFramework="4.5" />
    ...
</system.web>
...

```

Example calls:

```

PS python-crypto> python .\aspnetcrypto.py -entype owinauth -skey
BE5CF08F3D2E21DB3601E280503BF78E4EBD02D49245DCB37057DD1369A5172B sIqR0-
eLDTb5LBvpH54dU4LI-qPIF4a5EirVltpf7FEPWVnKsyh6-djZWag2_fs5a70ietPN0-
_DmfQKJrYSeGbbj f5Dt5CqWscqgKQSCjvBDev0EKUW4TS0Zm8VJA58rLpE877pybvFy_EifvdK8Dk_zIZYRPhNNHH
HffDtBqmD2ocIv6NkY4NkvEtmbRK04c7oLQMM-92LMAQWk-
SfCoRTUeWlj0PNrd6eQ5XZ97uwuVi6smGJ0uXoYAv_eFNjJkNg1EL82VKu2t4AMvxGgf1T6YAdC4pJvl9zlb8ew07
vVa6tSzcPrpe-KW2FhHU0HHnFUh7K0kloG-
Wwv_ePvKv02jusSWHFnb13f7zK7eksbnVncoT0J_Gce9wELMa3aBM3u56cIKViVhIQAzWg6nFRTsUh8LIxHXV0nFhk
7-3jndshd-QDv_KJ1C6rEyjIdgu61-2n2_jI-s3dt4fr70IG_U5dq6gGms3uXtLInEbXezTBxMW4RFGrqGafjMc-
B0SmGcCKXjskgbxtkZPyu2GiKTn0SncPMPv9bPP6d0I
02000000114170706c69636174696f6e436f6f6b6965010001000700000044687474703a2f2f736368656d617
32e786d6c736f61702e6f72672f77732f323030352f30352f6964656e746974792f636c61696d732f6e616d65
6964656e7469666965720a313734383430323039300100010001000873736f6c6e6963610100010001005
1687474703a2f2f736368656d61732e6d6963726f736f66742e636f6d2f616363657373636f6e74726f6c7365
72766963652f323031302f30372f636c61696d732f6964656e7469747970726f7669646572104153502e4e455
4204964656e746974790100010001001d4173704e65742e4964656e746974792e53656375726974795374616d
702464333830613932392d663538302d343038392d626263392d3535623238613439326538640100010001002
e687474703a2f2f6170706c69636174696f6e2f636c61696d732f617574686f72697a6174696f6e2f61637469
6f6e0e52573a5a616c6f67756a4a616b6f0100010001002e687474703a2f2f6170706c69636174696f6e2f636
c61696d732f617574686f72697a6174696f6e2f616374696f6e0e52503a5a616c6f67756a4a616b6f01000100
01001575726e3a64703a6c6f67676564757365727479706501490100010001000000000001000000020000000
72e6973737565641d5765642c203239204f637420323031342031323a31303a313720474d54082e6578706972
65731d5765642c203132204e6f7620323031342031323a31303a313720474d54
ApplicationCookie
Dhttp://schemas.xmlsoap.org/ws/2005/05/identity/claims/nameidentifier
1748402090 ssolnica
Qhttp://schemas.microsoft.com/accesscontrolservice/2010/07/claims/identityproviderASP.NET
IdentityAspNet.Identity.SecurityStamp$d380a929-f580-4089-bbc9-55b28a492e8d
.http://application/claims/authorization/actionRW:ZalogujJako
.http://application/claims/authorization/actionRP:ZalogujJako
urn:dp:loggedusertypeI.issuedWed, 29 Oct 2014 12:10:17
GM.expiresWed, 12 Nov 2014 12:10:17 GMT

PS python-crypto> python .\aspnetcrypto.py -entype antiforgery -skey
22d14047d53135334cb08d4b4d7da1dcfccd0eae9e66fea0b8dfdcda085a683
22d14047d53135334cb08d4b4d7da1dcfccd0eae9e66fea0b8dfdcda085a683 IgoPD31Z0v-eCQyBZeu_wL-
Vvr8IlmIpci-9iZD6F1S5Tf_HZb0G0MLE0jWU5aGjg_UtpYZ07vB3TRsrBEsJoTa5k4U1ygm68CmuBMAQ5G01
013382f4b0bc9f19dbce20a58893b4e32801 3é~Jč  ałô|N(

```

While writing this script I've learnt few interesting facts about encryption in ASP.NET. The keys you provide in the machineKey section are not directly used in encryption and validation logic, but **derivative keys** are created taking into account a context and a label (according to the [NIST specification](#)). A context in ASP.NET applications is a string: *User.MachineKey.Protect*. Label is different for each part of the framework. For ASP.NET

Identity cookies it's equal to

>*Microsoft.Owin.Security.Cookies.CookieAuthenticationMiddleware\1ApplicationCookie\102v1* when for Anti-Forgery token it's */System.Web.Helpers.AntiXsrf.AntiForgeryToken.v1*.

Another value would be used for forms authentication cookies. You may find some of those values in [the Purpose class source code](#). Another interesting fact is that **base64 url encoding** implementation differs between parts of the ASP.NET framework. Anti-Forgery tokens are encoded with one additional char specifying the number of padding characters ('=') when ASP.NET Identity cookies do not contain such information (padding characters number is calculated based on cookie value length modulo 4).

If you ever need to go deeper into ASP.NET cryptography, some interesting classes to look into are: [MachineKey](#), [AspNetCryptoServiceProvider](#), [NetFXCryptoService](#), [Purpose](#) and [SP800_108](#)

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